

Riemann Roch for Curves: Commutative Algebra I

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Preliminaries on Commutative Algebra

The Goal of these notes is to establish some basic facts from commutative algebra. In particular there are 3 important results we prove in these notes

1. The Noether Normalization for arbitrary fields. [0.19](#)
2. A generalised Nullstellensatz for arbitrary fields [0.26](#) which reduces to the conventional one over algebraically closed fields [0.30](#).
3. Finally we apply Noether Normalization to prove that for a finitely generated domain over a field, then the transcendence degree is the same as the Krull dimension [0.34](#).

However we do not prove the Cohen-Seidenberg theorems, since proving them would take us too far afield, although we do state the result that integral extensions of rings have the same dimension.

0.1 Generalities

All rings are commutative, unital and associative. For this we will denote by k an arbitrary field and by $\bar{k}$ an algebraic closure of k .

Definition 0.1. Let $\phi : A \rightarrow B$ be a ring homomorphism. In this case we say that B is an Algebra over A . Note that the algebra structure of B depends on the structure morphism $\phi : A \rightarrow B$. If $\psi : A \rightarrow C$ is another algebra over A , then a *morphism of A -Algebras* is a ring homomorphism $\eta : B \rightarrow C$ so that $\eta \circ \phi = \psi$ ¹.

Remark 0.2. We will often abuse notation and not write the structure morphism $\phi : A \rightarrow B$. In this case we will identify A with its image $\phi(A)$. This is slightly tricky and doesn't mean that the structure morphism is injective. Note that for any non-zero ideal $J \subset A$, we have that A/J is an A algebra with the canonical projection.

Definition 0.3. An A -Algebra B is called finitely generated or *finite type* if there is a finite set $\{b_1, \dots, b_n\} \subset B$ such that $B = A[b_1, \dots, b_n]$ i.e. any element in B can be written as a polynomial in the b_i 's.

Example 0.4. Let B be a finitely generated A algebra. Then choose a finite generating set $\{b_1, \dots, b_n\}$ and note that there is a short exact sequence

$$0 \longrightarrow I \longrightarrow A[x_1, \dots, x_n] \longrightarrow B \longrightarrow 0$$

where $A[x_1, \dots, x_n]$ is the polynomial ring in n variables and the morphism sends $x_i \mapsto b_i$. Thus we get an isomorphism $A[x_1, \dots, x_n]/I \cong B$. In this sense choosing a set of generators for B is equivalent to choosing coordinates.

Definition 0.5. An A -module M is called finitely generated or *finite* if there is a finite set $\{m_1, \dots, m_n\}$ such that $M = A \cdot m_1 + A \cdot m_2 + \dots + A \cdot m_n$. Observe that this is not a direct sum.

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¹i.e. it agrees on the image of A .

Example 0.6. Let M be a finite A module and choose a finite set of generators $\{m_1, \dots, m_n\}$ so that there is a short exact sequence

$$0 \longrightarrow I \longrightarrow A^{\oplus n} \longrightarrow M \longrightarrow 0$$

. where the middle term is the free A module on n generators. The morphism sends the 1 in the i^{th} coordinate to m_i . Thus we have an isomorphism $A^{\oplus n}/I \cong M$. Thus choosing generators is equivalent to choosing coordinates.

Remark 0.7. An A algebra B which is finite as an A module is necessarily finite type as an A algebra. However the other way is not true. Example if k is a field then $k[x]$ is a finite type k algebra. It has only one generator. However as a k module $k[x] = \bigoplus_n k \cdot x^n$.

Definition 0.8. Let $A \subset B$ be rings. Then an element $b \in B$ is called integral over A if b satisfies there is a monic polynomial $p(x) \in A[x]$ such that $p(b) = 0$. In other words there is an equation

$$b^n + a_1 b^{n-1} + \dots + a_n = 0$$

. We say that B is integral over A if the above condition is true for every $b \in B$.

Example 0.9. Consider $\mathbf{Z} \subset \mathbf{Q}$ the only elements of $\mathbf{Q}$ integral over $\mathbf{Z}$ are the integers. Indeed note that if $a/b \in \mathbf{Q}$ is any rational number with $\gcd(a, b) = 1$ integral over $\mathbf{Z}$ then we have that $(a/b)^n + (a/b)^{n-1}c_1 + \dots + c_n = 0$, then we have the following $-a^n = b(b^{n-1}c_n + \dots + a^{n-1}bc_1)$ which shows that b divides $a^n \implies b = \pm 1$. It is easy to see that the same hold in the case of any UFD considered as a subring of its fraction field.

0.2 Morphisms

Now following Grothendieck, it is forbidden to speak of objects having properties, but rather of relativising the whole circus and to speak of morphisms having the said properties. A general reference for this section is [AM69].

In this case we have another definition.

Definition 0.10. Let $\phi : A \rightarrow B$ be a morphism of rings, we say that

1. ϕ is of *finite type* if ϕ makes B into a finitely generated A algebra.
2. ϕ is *finite* if B becomes a finitely generated A module via the action $a \cdot b = \phi(a)b$.
3. ϕ is *integral* if B is integral over the subring $\phi(A)$.

Remark 0.11. A finite morphism is finite type. See remark 0.7.

Lemma 0.12. A finite morphism is integral.

Proof. Let $\phi : A \rightarrow B$ be a finite morphism. We identify A with $\phi(A)$. Now suppose $b_1, \dots, b_n$ generate B as an A module. Then for any $b \in B$ we have $b \cdot b_i = \sum_j a_{i,j} b_j$. Recasting this in determinant form we have

$$(b \cdot I - A) \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = 0$$

. Recall the classical adjoint of a matrix², and multiplying the left hand side of the equation by that matrix we get

$$\det(b \cdot I - A) \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = 0$$

. This implies $\det(bI - A)$ annihilates the whole ring B . This implies that it must identically be the zero endomorphism of the ring. Expanding we get a monic equation in b , which is the characteristic polynomial of A . □

²The classical Adjoint of a matrix A is the matrix $\text{Adj}(A)$ which satisfies the relation $\det(A) \cdot A^{-1} = \text{Adj}(A)$

Remark 0.13. The moral of 0.12 is this- whatever you can prove for finite you should try to prove for integral. In fact we will take my own advice and prove Noether's normalization as a statement about integral extensions following [Mum88].

Lemma 0.14. *Finite type and integral is finite.*

Proof. Suppose we find a finite generating set $\{b_1, \dots, b_n\}$ for B as a A algebra. Then note that each of these elements satisfy a monic equation with coefficients in A . Then if m is the largest degree of the monic polynomial satisfied by the b_i 's, then the set $\{b_1^{m_1} \dots b_n^{m_n} | 0 < m_1 + \dots m_n \leq n \cdot m\}$ does the trick. $\square$

Proposition 0.15. *Let $A \subset B$ be rings. The set $C := \{b \in B | b \text{ is integral over } A\}$ is a subring of B . This ring is the integral closure of A in B .*

Proof. Note that $1_B \in A$ and thus we can check that if b_1 and b_2 are in C then so is $b_1 - b_2$ and $b_1 b_2$.

Note that the subring $A[b_1, b_2]$ is finitely generated. However the generators are integral and therefore the inclusion $A \hookrightarrow A[b_1, b_2]$ is finite and thus integral by 0.12. $\square$

Remark 0.16. The same criteria works for a morphism of ring $\phi : A \rightarrow B$. We call that set $C := \{b \in B | b \text{ is integral over } \phi(A)\}$, the integral closure of A in C .

Proposition 0.17. *Let $\phi : A \rightarrow B$ and $\psi : B \rightarrow C$ be ring homomorphisms which are both integral (respectively finite or finite type), then their composition is integral (respectively finite or finite type).*

Proof. The non obvious case is that of the integral morphisms. So then let $c \in C$ be given. Then since C is integral over B , there is an equation

$$c^n + b_1 c^{n-1} + \dots + b_n = 0$$

Then note that $A[b_1, \dots, b_n]$ is integral over A and finite type by 0.14. It is therefore finite. Now the ring $A[b_1, \dots, b_n][c]$ is integral and finite type over $A[b_1, \dots, b_n]$ and therefore finite again by 0.14. Since the composition of two finite morphisms is finite and therefore integral by 0.7. $\square$

0.3 Applications to The Nullstellensatz.

In this section we develop some technology to prove the Nullstellensatz for arbitrary fields. We then deduce the Conventional Nullstellensatz for algebraically closed field. Our reference here is [Mum88] for the generalized Noether Normalization and [GW10] for the generalised Nullstellensatz.

Proposition 0.18. *Let $\phi : A \rightarrow B$ be an integral injective morphism of domains, then A is a field if and only if B is a field.*

Proof. Suppose A is a field, and choose $b \in B$. Since b is integral, $A[b]$ (the polynomial ring in b with coefficients in A) is a finite dimensional vector space. Now multiplication by b is an A linear injection. It is thus a bijection. Therefore b is a unit.

Now suppose B is a field. Then $a \in A$ and $a^{-1} = c$ lives in B . By integrality there is an equation $c^n + a_1 c^{n-1} + \dots + a_n = 0$ where the coefficients live in A . Now we have an equation $c^n = -(a_1 c^{n-1} + \dots + a_n)$. Multiplying throughout by a^{n-1} we get $a^{-1} = -(a_1 + \dots + a_n a^{n-1})$. Note that the right side lives in A . We are done. $\square$

Theorem 0.19 (Noether Normalization). *Let A be a finitely generated domain over a field k . Let $K = \text{Frac}(A)$. Let $n = \text{tr.deg}(K/k)$, then there exist algebraically independent elements $x_1, \dots, x_n \subset A$ and a subring $B \subset A$ such*

1. B is isomorphic as a k -algebra to $k[x_1, \dots, x_n]$.
2. The inclusion $k[x_1, \dots, x_n] \hookrightarrow A$ is integral.

Before we prove this it is advisable to begin with some preparation.

Lemma 0.20. *Let A be a finitely generated domain over a (not necessarily algebraically closed field) k and let $K = \text{Frac}(A)$. Let $n = \text{tr.deg}(K/k)$. Let m be the cardinality of a generating set of A as a k -Algebra. Then $m \geq n$.*

Proof. Choosing m generators for A we get $A \cong k[x_1, \dots, x_m]/I$. Now every element in the A can be written as a polynomial in the x_i modulo some relations. In the fraction field K one has that every element is the solution of the equation $ax - b$ as $a, b \in A$. Thus every element in K is algebraically dependent on the subring generated by the x_i and so the transcendence degree of K cannot be larger than m . Also note that if the transcendence degree was exactly m , then the ideal $I = 0$ because otherwise there would be a non-trivial relation among the generators in the x_i forcing the transcendence degree to be less than m . $\square$

Proof. (of 0.19) Using the fact that A is a domain and 0.4 we see that $A \cong k[y_1, \dots, y_m]/I$ where I is a prime ideal. Now note that we see that $\text{tr.deg}(K/k) = n \leq m$ by 0.20. If $m = n$ then we are done once again by 0.20.

So suppose $m > n$, then $I \neq 0$ and in particular there is a polynomial $f(y_1, \dots, y_m) = 0$. We will now use induction on m to prove the statement. In particular we will show that there is a subring $B = k[z_2, \dots, z_m]$ inside of A over which A is integral. By induction there is a subring $k[x_1, \dots, x_n]$ inside of B over which it is integral. Since the composition of two integral morphisms is integral we will be done.

So now there is a polynomial $f(y_1, \dots, y_m) = 0$. Now we may break the polynomial terms as $f = f_1 + \dots + f_k$ where each f_j is a monomial term. Now each monomial f_j looks like

$$f_j = b_j \prod_i y_i^{a_{j,i}}$$

where $b_j \in k$ and $a_{j,i}$ are non negative integers.

We now choose non-negative integers $r_1, \dots, r_n$ so that $z_1 = y_1$ and $z_n = y_i - y_1^{r_i}$ for all $i > 1$. Then we have the equation

$$f(y_1, y_1^{r_2} + z_2, \dots, y_1^{r_n} + z_n) = 0$$

Doing the change of coordinates as above, each monomial term f_j will end up expanding out, and give us an equation of the form

$$b_j \prod_i y_i^{a_{j,i}} = b_j y_1^{a_{j,1}r_1 + \dots + a_{j,n}r_n} + \text{mixed terms}$$

Depending upon the $a_{i,j}$ we may choose our integers r_i so that the degree of the pure y_1 term in each monomial term f_j is different. This is always possible. Finally we may choose each r_i to be very large so that the following equation holds

$$f(y_1, y_1^{r_2} + z_2, \dots, y_1^{r_n} + z_n) = cy_1^N + \text{lower degree terms} = 0$$

But since $c \in k^*$ we can cancel it out of the leading term and we get an integral equation for y_1 in terms of the z_i . Thus y_1 is integral over the ring $k[z_2, \dots, z_n]$. By induction we are done. $\square$

Corollary 0.21. (Zariski's Lemma) *If k is a field, A is a finitely generated k -algebra, and $K = A/\mathfrak{m}$ where $\mathfrak{m}$ is a maximal ideal in A , then K is a finite extension of k .*

Proof. Notice that K is a field which is finitely generated as a k algebra and note that its fraction field is itself. Then if $n = \text{trdeg}(K/k)$, then there is an integral morphism $k[x_1, \dots, x_n] \hookrightarrow K$, but then this implies $k[x_1, \dots, x_n]$ is a field (by 0.18), which then shows us that $n = 0$ (Indeed, if x_1 was invertible then $x_1 \cdot f = 1$, then by evaluating at 0 we get $0 = 1$, a contradiction). Thus K is finite type and integral over k . This implies that it is finite over k . (cf. 0.14.) $\square$

Corollary 0.22. (Weak Nullstellensatz) *Every maximal ideal of $\bar{k}[x_1, \dots, x_n]$ is of the form $(x_1 - a_1, \dots, x_n - a_n)$ for some $(a_1, \dots, a_n) \in \bar{k}^n$.*

Proof. Let $\mathfrak{m}$ be a maximal ideal of $\bar{k}[x_1, \dots, x_n]$, then $\bar{k}[x_1, \dots, x_n]/\mathfrak{m}$ is a finite extension of $\bar{k}$ by 0.21 which implies that it is $\bar{k}$. Let $a_1, \dots, a_n$ be the image of $x_1, \dots, x_n$ then $(x_1 - a_1, \dots, x_n - a_n) \subset \mathfrak{m}$ which by maximality implies equality. $\square$

Remark 0.23. The weak form of the Nullstellensatz is actually quite important, it gives us a way of identifying the points of $\mathbf{A}_k^n$ with maximal ideals of $\bar{k}[x_1, \dots, x_n]$.

Now we will prove a general version of Hilbert Nullstellensatz over arbitrary fields, which will reduce to the well known case if the field is algebraically closed.

Lemma 0.24. *Let A be a finitely generated k algebra, and L a finite extension of k . For any morphism of k algebras $\phi : A \rightarrow L$, the kernel of ϕ is maximal in A .*

Proof. Note that $\phi(A)$ is a subring of L and thus a domain. However it is also finite as a module over k . By 0.7 and 0.18, $\phi(A)$ is a field. Thus the kernel of ϕ is maximal in A . $\square$

Recall that the Jacobson radical of a ring is the intersection of all maximal ideals of that ring.

Lemma 0.25. *Let A be an integral domain finitely generated as an algebra over a field k . Then the Jacobson radical is 0.*

Proof. Suppose not then there is an element $x \neq 0$ which belongs to the Jacobson radical of A . Note that x cannot be a unit. Thus we can invert x to get the ring $A[x^{-1}]$. The morphism $A \rightarrow A[x^{-1}]$ is an inclusion. Now note that if $\mathfrak{m}$ is maximal in $A[x^{-1}]$ then by 0.21 $A[x^{-1}]/\mathfrak{m}$ is a finite extension of k . Thus using 0.24 the canonical map $\phi : A \hookrightarrow A[x^{-1}] \rightarrow A[x^{-1}]/\mathfrak{m}$ has $\ker(\phi) \subset A$ a maximal ideal. However $x \notin \ker(\phi)$ since it is a unit in $A[x^{-1}]$ and the map $A \rightarrow A[x^{-1}]$ is an injection. Thus we have a contradiction. $\square$

Theorem 0.26. (Hilbert's Nullstellensatz) *Let A be a finitely generated algebra over a (not necessarily algebraically closed) field k . Then the ring is Jacobson. That is for every prime ideal $\mathfrak{p} \subset A$ we have that*

$$\mathfrak{p} = \bigcap_{\mathfrak{m} \supset \mathfrak{p}} \mathfrak{m}$$

Proof. Given any prime $\mathfrak{p} \subset A$ observe that $A/\mathfrak{p}$ is a domain. The maximal ideals are in bijection with maximal ideals containing $\mathfrak{p}$. Thus the proof is equivalent to the assertion that the Jacobson radical of $A/\mathfrak{p}$ is 0. But this follows from 0.25. $\square$

Lemma 0.27. *Let $I \subset \bar{k}[x_1, \dots, x_n]$. Then there is a bijection between all maximal ideals containing I and the vanishing locus $V(I)$.*

Proof. From 0.22 every maximal ideal of $\bar{k}[x_1, \dots, x_n]$ is of the form $(x_1 - a_1, \dots, x_n - a_n)$. Now suppose $I \subset (x_1 - a_1, \dots, x_n - a_n)$ then we see that $f \in I \implies f \in (x_1 - a_1, \dots, x_n - a_n)$. This means that $f = \sum_{i=1}^n p_i(x_i - a_i)$. Clearly then $f(a_1, \dots, a_n) = 0$. Now pick a point $p = (a_1, \dots, a_n) \in V(I)$ then let $f \in I$. Consider the evaluation morphism $ev_p : k[x_1, \dots, x_n] \rightarrow k$ which evaluates every polynomial at $p = (a_1, \dots, a_n)$. Then by 0.22 this is the ideal $(x_1 - a_1, \dots, x_n - a_n)$ and we see that $f \in (x_1 - a_1, \dots, x_n - a_n)$. $\square$

Remark 0.28. By 0.27 we can say that a polynomial with coefficients in $\bar{k}$ vanishes at a point p if and only if it is contained in the maximal ideal corresponding to that point. Thus the ideal of a set in S inside $\mathbf{A}_k^n$ is equivalently defined as $I(S) = \bigcap_{p \in S} \mathfrak{m}_p$, where the intersection is taken over all maximal ideals corresponding to the points $p \in S$.

Remark 0.29. Recall that for an ideal I , its radical $\sqrt{I}$, can be written as the intersection $\bigcap_{I \subset \mathfrak{p}} \mathfrak{p}$. If the ring is Jacobson i.e. each prime ideal is the intersection of all maximal ideals containing it, then $\sqrt{I} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$. To see this first note that since maximal is prime $\bigcap_{I \subset \mathfrak{p}} \mathfrak{p} \subset \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$. Now however since each prime is the intersection of all maximal ideal containing it if x lives in all maximal ideals containing I , it must live in each prime ideal containing I . Hence $\bigcap_{I \subset \mathfrak{p}} \mathfrak{p} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$

Proposition 0.30. (Conventional Nullstellensatz) *Let J be an ideal in $\bar{k}[x_1, \dots, x_n]$, then we have $IV(J) = \sqrt{J}$.*

Proof. Getting together everything noted in 0.27, 0.28 and 0.29 we get the following sequence of equalities-

$$IV(J) = \bigcap_{p \in V(J)} \mathfrak{m}_p = \bigcap_{J \subset \mathfrak{m}_p} \mathfrak{m}_p = \bigcap_{J \subset \mathfrak{p}} \mathfrak{p} = \sqrt{J}$$

$\square$

0.4 Applications to Dimension Theory

The goal in this section is to prove that for an integral finitely generated k algebra the Krull dimension is the same as the transcendence degree.

Definition 0.31. Let R be a ring. The Krull Dimension of R , denoted $\dim(R)$ is the (possible infinite) number

$$\dim(R) = \sup\{ l \in \mathbb{N} \mid 0 \subsetneq \mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_l, \text{ where each } \mathfrak{p}_i \text{ is a prime ideal in } R \}$$

Example 0.32. Consider the ring $k[x_1, x_2, x_3, \dots]$ which is the polynomial ring in infinitely many variables. Then this ring has infinite Krull Dimension. Of course this ring is not Noetherian. However there exist Noetherian ring of infinite Krull dimension.³

Proposition 0.33. Let $A \subset B$ be an integral extension. Then

$$\dim(A) = \dim(B)$$

Proof. This theorem is immediate from the Cohen-Seidenberg going up and going down theorems(cf. [AM69]). However stating and proving them will take us too far afield. We may write down the whole proof if at a later point we feel that it serves a purpose. $\square$

Proposition 0.34. Let R be an integral domain finitely generated as a k -Algebra. Let $K = \text{Frac}(R)$. Then $\dim(R) = \text{trdeg}(K/k)$.

Proof. If we set $n = \text{trdeg}(K/k)$. Then applying Noether Normalization we have $k[x_1, \dots, x_n]$ a subring over which R is integral. Then $\dim(k[x_1, \dots, x_n]) = \dim(R) = n$. $\square$

References

- [AM69] M.F. Atiyah and I.G. MacDonald, Introduction to Commutative Algebra, Addison-Wesley Publishing Company, 1969
- [Mum88] D. Mumford, The red book of varieties and schemes, Lecture Notes in Mathematics, 1358, Springer-Verlag, Berlin, 1988
- [GW10] U. Görtz and T. Wedhorn. Algebraic Geometry 1: Schemes, Vieweg + Teubner Verlag — Springer Fachmedien Wiesbaden GmbH 2010

³google it if you are curious.